

# Point-Wise Contraction-Based Control Based on Vector-Space Optimization under State and input Constraints

Myeongseok Ryu, Kyunghwan Choi and Sesun You

*Abstract*—Contraction theory offers

## I. INTRODUCTION

- Advantages of contraction-based control
- Challenges in existing methods
  - Absence of explicit handling of constraints
  - Requirements of prior computation of contraction metrics
  - Limitations to online optimization in real-time applications (computational complexity)
- Potential of vector-space optimization
  - Integration of constraints directly into the optimization framework
  - Input constraints and state constraints can be handled explicitly
  - CBF based constraints can be easily integrated
- Justification of the proposed method
  - Examples of effectiveness of point-wise control
  - Examples of point-wise reference

## II. PRELIMINARIES

## III. MAIN RESULTS

- Constraint vs Stability analysis

## IV. NUMERICAL VALIDATIONS

### A. Validation Setup

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$$\frac{d}{dt} \begin{pmatrix} \beta \\ r \\ \theta_f \\ \theta_r \end{pmatrix} = \begin{pmatrix} \frac{1}{mv_x} (F_f + F_r) - r \\ \frac{1}{I_z} (l_f F_f - l_r F_r) \\ \frac{1}{\tau} (\omega_f - \theta_f) \\ \frac{1}{\tau} (\omega_r - \theta_r) \end{pmatrix}, \quad (1)$$

where the lateral tire forces  $F_f$  and  $F_r$  are simplified as

$$F_f = F_{\max} \tanh\left(\frac{C_f}{F_{\max}} \alpha_f\right), F_r = F_{\max} \tanh\left(\frac{C_r}{F_{\max}} \alpha_r\right), \quad (2)$$

with the slip angles  $\alpha_f$  and  $\alpha_r$  defined as

$$\alpha_f(\beta, r, \theta_f) = \beta + \frac{l_f r}{v_x} - \theta_f, \alpha_r(\beta, r, \theta_r) = \beta - \frac{l_r r}{v_x} - \theta_r. \quad (3)$$

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M. Ryu and K. Choi are with Cho Chun Shik Graduate School of Mobility, Korea Advanced Institute of Science and Technology (KAIST), 193 Daejeon, Republic of Korea {dding\_98, kh.choi}@kaist.ac.kr

P. Misra is with the Department of Electrical Engineering, Wright State University, Dayton, OH 45435, USA pmisra@cs.wright.edu

TABLE I: Vehicle Dynamics Parameters.

| Symbol       | Description                        | Value                                                       |
|--------------|------------------------------------|-------------------------------------------------------------|
| $m$          | Vehicle Mass                       | 1650 kg                                                     |
| $(l_f, l_r)$ | Front and Rear Wheel-base          | (1.11, 1.59) m                                              |
| $(C_f, C_r)$ | Front and Rear Cornering Stiffness | (133, 98.8) kN rad <sup>-1</sup>                            |
| $F_{\max}$   | Maximum Tire Force                 | 8000 N                                                      |
| $I_z$        | Yaw Moment of Inertia              | 2315.3 kg m <sup>2</sup>                                    |
| $v_x$        | Longitudinal Velocity              | 27.7 m s <sup>-1</sup> ( $\approx 100$ km h <sup>-1</sup> ) |

By denoting the state vector as  $\mathbf{x} = (\beta, r, u_f, u_r)^\top \in \mathbb{R}^4$  and the control input as  $\mathbf{u} = (w_f, w_r) \in \mathbb{R}^2$ , the system can be represented in the SDC formulation as

$$\frac{d}{dt} \mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad (4)$$

where  $\mathbf{x} = (\beta, r, u_f, u_r)^\top \in \mathbb{R}^4$  is the state vector,

Following constraints are considered for the validation:

*State Constraints:* To prevent tire slip, the slip angles  $\alpha_f$  and  $\alpha_r$  are constrained as

$$|\alpha_f| \leq \alpha_{f,\max}, \quad |\alpha_r| \leq \alpha_{r,\max}, \quad (5)$$

*Input Constraints:* Considering physical limitations of the steering system, the control inputs, steering angle rate, are constrained as

$$|w_f| \leq w_{f,\max}, \quad |w_r| \leq w_{r,\max}. \quad (6)$$

### B. Controllers for Comparison

We compare the following controllers including the proposed controller as follows:

*Proposed Method:*

- Vector-space optimization
- CBF + Input saturation

*CV-STEM:*

*Desired control law only:*

### C. Validation Results

## V. CONCLUSIONS

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